

1.1 Overview

The advancement of modern electronic technologies has led to a continuous increase in device performance, miniaturization, and functional integration. As a result, the heat generated per unit area in components such as central processing units (CPUs), power transistors, and laser diodes has risen significantly. Efficient thermal management is therefore one of the most critical challenges in ensuring the stability, performance, and reliability of electronic systems. Excessive heat can cause thermal stress, performance degradation, or even permanent failure of electronic components.

Traditional air-cooling systems using fans and extended surface heat sinks are simple and inexpensive, but they are unable to handle the extremely high heat fluxes associated with modern high-performance devices. Air has a low thermal conductivity, which limits its heat dissipation capability. Consequently, there has been a growing demand for compact, high-efficiency cooling solutions capable of removing large amounts of heat from confined areas. One such advanced method is liquid cooling using minichannel heat sinks.

1.2 Concept of Minichannel Heat Sink

The concept of the minichannel heat sink (MCHS) was first introduced by Tuckerman and Pease in 1981, who demonstrated that minichannels could remove heat fluxes up to 790 W/cm^2 from silicon chips. Since then, MCHSs have become one of the most promising technologies for high-power-density cooling applications. A minichannel heat sink consists of a solid substrate typically made of copper, aluminium, or silicon containing small parallel channels with hydraulic diameters less than 1 mm. A coolant such as water, glycol, or nanofluid flows through these channels, carrying heat away from the device surface through forced convection.

Minichannel heat sinks are attractive because they offer a very high surface-area-to-volume ratio, which greatly enhances heat transfer applications. However, their design involves a balance between heat transfer performance and hydraulic resistance, as the small channel sizes often result in a higher pressure drop requiring greater pumping power.

In this project, a Minichannel heat sink is designed, model, and analysed using computational fluid dynamics (CFD) to study its thermal and fluid performance under different operating conditions. Parameters such as channel geometry, flow rate, coolant type, and material are investigated to identify an optimal configuration that ensures maximum heat transfer with minimum energy consumption. The goal is to develop an efficient and compact liquid-cooling solution that can meet the growing heat dissipation demands of advanced electronic systems.

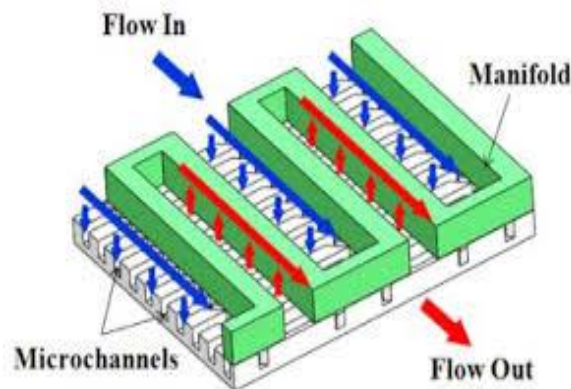


Fig 1.1: Minichannel Heat Sink

1.3 Project Overview

The project “Minichannel Heat Sink for Electronics Cooling” focuses on the design and simulation of a compact liquid-cooled heat sink capable of effectively removing high heat flux from electronic components. The study primarily involves numerical modelling using ANSYS Fluent to analyse temperature distribution, pressure drop, and heat transfer behaviour within the Minichannels. Minichannel heat sinks address this need by utilizing microscale liquid flow to achieve high heat removal rates in compact structures.

In this project, a rectangular Minichannel heat sink is model with multiple parallel channels. The working fluid (coolant) flows through the channels, absorbing heat from the base surface that simulates the heat generated by an electronic chip. Various coolants, such as water, ethylene glycol, and nanofluids, are analysed for their thermal performance. The design parameters such as channel width, height, length, and number of channels are optimized to achieve the best balance between heat transfer rate and pumping power.

CFD simulations are used to predict flow behaviour, temperature contours, and pressure drop across the Minichannels. The results provide valuable insights into how geometrical modifications and fluid properties influence overall heat sink performance. The outcomes of this project are applicable to a wide range of high-power electronic systems such as CPUs, GPUs, power modules, and laser diode arrays.

1.4 Industrial Applications of Minichannel Heat Sink

A minichannel heat sink is a passive or active cooling device that absorbs and dissipates heat from electronic components to the surrounding air or coolant, thereby maintaining safe operating temperatures. Minichannel heat sinks (also called mini-channel or micro-channel heat sinks, depending on size) are widely used in industries where high heat removal, compact size, and efficiency are required. The major industrial applications explained clearly below:

- **Power Electronics Industry:** Used for high power electronic systems where heat generation is continuous and intense.
- **Electronics and Semiconductor Industry:** Minichannel heat sinks are extensively used in the electronics and semiconductor industry due to the continuous increase in power density of electronic components. Modern processors such as CPUs, GPUs, and ASICs generate very high heat fluxes in a compact area.

- **Automotive Industry:** In the automotive sector, especially with the growth of electric and hybrid vehicles, minichannel heat sinks play a vital role in thermal management systems. They are used for cooling battery packs, where maintaining a uniform temperature is critical for battery safety, performance, and longevity.
- **Aerospace and Defence:** Aerospace and defence systems require highly reliable and lightweight thermal management solutions, making minichannel heat sinks ideal. They are used to cool avionics, navigation systems, flight control electronics, and onboard computers, which operate continuously under harsh environmental conditions.

1.5 Conclusion

The “Minichannel Heat Sink for Electronics Cooling” project presents an innovative and efficient approach to managing the growing thermal challenges in modern electronic systems. By utilizing micro-scale liquid cooling, it achieves a high rate of heat removal in a compact and energy-efficient manner. The use of CFD simulation provides deep insights into the thermal and hydraulic behaviour of Minichannel structures, allowing for design optimization without extensive physical testing.

This project demonstrates that Minichannel heat sinks can significantly outperform conventional cooling systems, offering a practical solution for next-generation electronics that demand high performance and miniaturization. The knowledge gained from this study can serve as a foundation for future research and industrial applications in the field of advanced thermal management.

2.1 REVIEW OF LITERATURE

Gilmore et.al [1] proposed an open manifold Minichannel heat sink designed to improve cooling performance for high heat flux electronic devices while reducing pressure drop. The study used detailed three-dimensional conjugate heat transfer models to compare the open design with a conventional manifold Minichannel geometry. By removing channel walls beneath the slot nozzles, the new design achieved up to a 75% reduction in pressure drop with only a small change in thermal resistance. The researchers also identified that optimizing the manifold flow distribution could further improve performance. Their results demonstrated that efficient cooling can be achieved without increasing system complexity or size. This study is relevant to my project as it highlights effective design modifications and modelling approaches that enhance heat transfer while minimizing flow resistance in Minichannel cooling systems.

Ghasemi et.al [2] carried out a numerical investigation to improve the performance of a solar parabolic trough collector using an Al-water nanofluid. The main goal was to analyse how adding nanoparticles to the working fluid affects heat transfer and overall system efficiency. The researchers used computational fluid dynamics (CFD) modelling to simulate thermal behaviour under different flow and heat flux conditions. Their results showed that using nanofluids significantly enhanced the collector's heat transfer rate compared to conventional fluids, leading to better thermal performance. The study concluded that nanofluids can effectively improve energy conversion in solar thermal systems. This research supports my project by demonstrating the importance of fluid properties and thermal optimization methods, which are directly relevant to enhancing heat transfer in compact cooling systems.

Xu et.al [3] carried out a numerical study on the thermal and fluid performance of micro-channel heat sinks with different configurations. The main aim of their

research was to enhance the heat transfer rate while maintaining a reasonable pressure drop for electronic cooling systems. They used computational fluid dynamics (CFD) simulations to examine how changes in channel shape, aspect ratio, and flow rate affect temperature distribution and fluid flow. The results showed that optimized channel geometries significantly improved heat transfer efficiency and reduced hot spots. The study concluded that proper design and geometric optimization can balance cooling performance and flow resistance. This work supports my project by providing useful guidance on how micro-channel structure and fluid dynamics can be optimized to improve heat dissipation in compact thermal systems.

Gong et.al [4] conducted a numerical study to improve the design of micro-channel heat sinks for electronic device cooling. The researchers modelled four types of heat sinks—traditional, pin-fin, single-jet, and double-layer—to compare their flow and heat transfer characteristics. Using computational simulations based on steady-state fluid and heat transfer equations, they analysed pressure drop, Nusselt number, and field coordination to assess performance. Results showed that adding metal foam to the inlet improved flow uniformity and heat transfer, while pin-fin and double-layer structures enhanced cooling at moderate pressure drops. The single-jet model achieved the highest cooling but suffered from excessive flow resistance. This study highlights the importance of micro-channel layout in thermal management and supports my project by providing valuable insights into optimizing heat sink geometry for efficient electronic cooling.

Goodarzi [5] studied the thermal performance of Minichannel heat sinks using graphene–silver/water hybrid nanofluid as the coolant. The aim of the work was to examine how nanoparticle concentration, fluid temperature, Reynolds number, and channel shape influence heat transfer. The authors conducted a numerical analysis on three Minichannel geometries: circular, trapezoidal, and square. The results showed that increasing nanoparticle concentration improved thermal conductivity and heat transfer, with the square channel providing the highest heat transfer enhancement. However, higher concentrations also increased viscosity, leading to

greater pressure drop. Overall, the study demonstrates that both working fluid selection and channel geometry strongly affect cooling efficiency. This is relevant to my project because it highlights how fluid properties and structural design can be optimized to improve heat dissipation performance.

Ayatollah et.al [6] investigated the performance of a newly designed flattened Minichannel heat sink for electronic cooling applications. The aim of their study was to reduce thermal resistance and improve temperature uniformity compared to conventional Minichannel heat sinks. They used a three-dimensional numerical simulation based on the finite volume method to analyse fluid flow and heat transfer under different geometric configurations. The study compared three designs and evaluated parameters such as number of channels, wall thickness, separation plate location, and velocity ratio. The results showed that the double-layer flattened design produced the lowest thermal resistance and most uniform temperature distribution, improving cooling performance by up to 7.9% compared to standard designs. This study is relevant to my project as it highlights how channel geometry and flow arrangement directly affect heat dissipation efficiency, supporting the need to optimize heat sink design for enhanced thermal management.

Arshad et.al [7] conducted an experimental study on heat transfer and pressure drop in a straight minichannel heat sink using a TiO_2 –water nanofluid. The study aimed to evaluate how nanofluids enhance cooling performance compared to distilled water under different heating powers (100 W, 125 W, 150 W). Using a copper heat sink and various Reynolds numbers, they measured temperature, pressure drop, and thermal resistance. The results showed that TiO_2 nanofluid improved heat transfer efficiency, especially at lower heating powers, achieving up to 12.75% higher Nusselt numbers and reduced wall temperatures. However, it also caused higher pressure drops due to increased viscosity. This work demonstrates that nanofluids can significantly enhance thermal management in compact heat sinks, which supports my project's focus on improving heat dissipation efficiency using advanced cooling fluids.

Saeed and Kim et. al [8] presented a study on header design approaches for mini-channel heat sinks using both analytical and numerical methods to improve cooling performance in electronic systems. The main goal was to reduce flow mal distribution and pressure drop caused by inefficient header geometries. The authors developed a mathematical model in MATLAB and validated it using ANSYS-CFX simulations. They analysed how geometric parameters affect coolant flow, pressure loss, and thermal resistance. Their findings showed that optimized header designs significantly reduced flow mal distribution by up to 74%, pressure drop by 71%, and thermal resistance by 5% compared to conventional designs. This research is relevant to my project as it demonstrates how careful design optimization of fluid distribution components can enhance the thermal and hydraulic performance of liquid-cooled heat sinks, contributing to more efficient and compact thermal management systems

Joshi [9] studied liquid cooling technologies for data centres to address the rising power densities and thermal challenges in high-performance computing systems. The research focused on improving energy efficiency and system reliability through advanced cooling designs. Using both simulation and experimental methods, the study analysed different liquid cooling configurations, including cold plates, manifolds, and coolant distribution systems. The results showed that liquid cooling significantly reduces processor temperatures and overall energy consumption compared to traditional air cooling. The authors also discussed design considerations like flow rate, material compatibility, and maintenance issues. This study supports my project by providing useful insights into efficient thermal management techniques and demonstrating the benefits of liquid cooling for enhancing system performance and sustainability in high-power computing environments.

Patel [10] investigated cooling strategies for high-performance data centres to reduce energy use and improve system reliability. The study focused on comparing air and liquid cooling systems under different thermal loads. Using experimental testing and computational modelling, the authors analysed heat transfer

performance, fluid flow, and temperature distribution in both methods. Their findings showed that liquid cooling provides higher thermal efficiency, lower operational costs, and better temperature uniformity than conventional air cooling. The research also discussed design factors such as coolant type, flow rate, and system layout that influence performance. This study supports my project by highlighting the effectiveness of liquid cooling in managing high heat densities and guiding design considerations for efficient thermal management in modern electronic or computing systems.

Gullbrand [11] from Intel Corporation studied liquid cooling technologies for high-performance compute systems to address the growing heat and power density in modern servers. The paper focused on the design and analysis of key liquid cooling components, including cooling distribution units (CDUs), manifolds, fluids, quick disconnects, and cold plates. Using computational fluid dynamics (CFD) and experimental analysis, the authors compared different cold plate designs and evaluated cooling performance for both high- and low-power components. Results showed that liquid cooling offers much higher heat removal efficiency than traditional air cooling and supports compact, high-density systems. The study also highlighted design challenges such as flow distribution, corrosion control, and serviceability. This research is relevant to my project as it provides detailed insights into liquid cooling design and performance optimization, supporting the development of efficient thermal management systems for CPUs and electronic devices.

Harun et.al [12] reviewed the development of liquid cooling systems for central processing units (CPUs) to address the increasing heat generation in modern compact electronics. Their study examined various cooling methods such as heat sinks, heat pipes, thermoelectric coolers, and jet impingement systems and compared the performance of different working fluids, including water, ethylene glycol, and nanofluids. The authors found that nanofluids, which contain dispersed nanoparticles, significantly enhance heat transfer due to improved thermal conductivity. They also highlighted that optimal design of liquid blocks and channel

configurations improves cooling efficiency and temperature uniformity. This review is relevant to my project as it provides a comprehensive understanding of how liquid and nanofluid cooling systems can improve thermal management, guiding the selection of materials and designs for efficient heat dissipation in high-performance processors.

Pal et.al [13] investigated liquid-cooled thermal management architectures for high-power electronics in "more electric" aircraft, aiming to address the significant thermal and packaging challenges posed by high heat fluxes. The study used system analysis and Computational Fluid Dynamics (CFD) modelling to assess the entire heat path, from acquisition to ultimate rejection. A key finding was that while liquid cooling efficiently manages internal heat, the system's overall performance is constrained by its reliance on external ram/fan air for heat rejection. This limitation dictates that, under worst-case hot day ground conditions, the coolant inlet temperature must be tolerated up to 70°C. This paper is essential to my project as it establishes the fundamental thermal boundary conditions and design requirements that must be met by any proposed next- generation cooling technology

Ramesh et.al [14] conducted a numerical investigation on improving the thermal performance of phase change material (PCM)-based hybrid Minichannel heat sinks (MCHS) for electronic cooling. The study aimed to evaluate how integrating PCM layers into Minichannel designs could enhance heat dissipation and temperature uniformity. Using ANSYS Fluent simulations, the authors model six different PCM-based heat sink configurations and compared their performance with a conventional design without PCM. Results showed that the model with PCM placed at the top layer achieved 7.3% lower thermal resistance and 15.26% better temperature uniformity than the standard design. The findings proved that PCM integration helps stabilize temperature fluctuations under varying heat loads. This study supports my project by highlighting how hybrid Minichannel designs and thermal storage materials can significantly improve electronic cooling efficiency and reliability.

2.2 Research Gap

Numerical investigations using CFD tools have shown that increasing coolant flow rate enhances heat transfer performance; however, it also leads to a higher pressure drop across the channel. Some researchers attempted geometric modifications such as varying channel aspect ratio, fin thickness, and channel spacing to improve thermal performance. Despite these improvements, many studies focused mainly on thermal enhancement without giving sufficient importance to pumping power and overall system efficiency.

1. **Limited Optimization of Channel Geometry:** Existing studies mostly analyse standard rectangular minichannels. Effects of channel depth, width ratio, fin thickness, and aspect ratio are not fully optimized together. Lack of comparative study between different minichannel geometries for electronics cooling
2. **Trade-off Between Heat Transfer and Pressure Drop:** Increasing heat transfer often leads to high pressure drop. Most studies prioritize thermal performance but ignore pumping power. Practical electronics cooling requires energy-efficient solutions
3. **Lack of Experimental Validation:** Many results are purely numerical Limited experimental data to validate CFD results. Manufacturing and surface roughness effects are often neglected. Need for experimental validation of minichannel heat sink performance.
4. **Working Fluids for Liquid CPU Cooling:** The working fluid is a critical component in liquid CPU cooling systems because it directly affects heat transfer efficiency, system reliability, and safety.

2.3 Objective of the Project

The objective of this project is to design, analyse, and evaluate the performance of a minichannel heat sink for effective electronics cooling. With the continuous

miniaturization of electronic devices and the increase in power density, efficient thermal management has become a critical requirement to ensure reliability and long service life of electronic components. This project aims to develop a compact minichannel heat sink capable of dissipating high heat flux while maintaining the operating temperature within safe limits. The study focuses on understanding the heat transfer and fluid flow characteristics of the minichannel heat sink under different operating conditions, such as varying heat flux and coolant flow rates. Special attention is given to analysing the pressure drop across the minichannels, as excessive pressure loss can increase pumping power and reduce overall system efficiency. The project also aims to investigate the effect of channel geometry parameters, including channel width, depth, spacing, and aspect ratio, on thermal performance. Numerical analysis using computational fluid dynamics (CFD) techniques is employed to predict temperature distribution, heat transfer coefficient, and flow behaviour inside the minichannels. The results obtained from the proposed design are compared with conventional heat sink configurations to evaluate performance enhancement. Overall, the objective of this project is to optimize the design of a minichannel heat sink that provides improved heat transfer performance with acceptable hydraulic characteristics, making it suitable for practical electronics cooling applications.

The methodology outlines the systematic approach used to achieve the project objectives. Here we describe the steps, tools, and techniques employed to design and analyse the Heat sink.

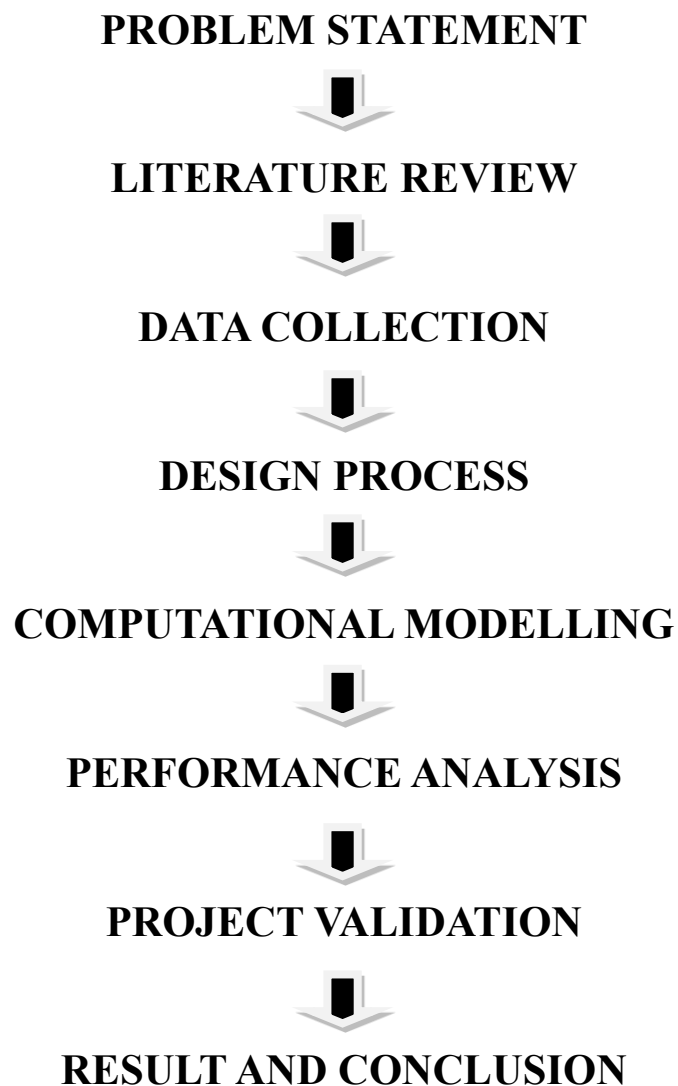


Fig 3.1 Flow Chart of Methodology

1. **Problem Statement:** The first step of our project was to identify and define the problem. And so, the objective of the project was decided.
2. **Literature Review:** After clearly defining the objective of the project, necessary research was conducted to study the existing literature, research papers, patents and technical documents related to the domain. A solid understanding of the current methodologies, analysis techniques and new trends were made.
3. **Data Collection:** Next step that we undertook was to collect the necessary data and specifications to design the heat exchanger.
4. **Design Process:** After getting the required data, the designing process was started. The components of the heat exchanger were designed and assembled in the SOLIDWORKS software.
5. **Computational Modelling:** After properly designing all the geometries, the designs were subjected to the analysis platform viz. ANSYS Fluent. The mesh generation was done, all the boundary conditions were defined, and solver process was defined.
6. **Performance Analysis:** After defining the required parameters, the results were obtained. The various results were then compared to give a final conclusion about the project. Common parameters were taken to compare the results.
7. **Project Validation:** The next step after getting the result and comparisons was to do a validation of the project by conducting real life experiments. Since doing the real-life experiment was not feasible in this case, so the validation of the structural analysis was done using a previous research paper titled “Numerical Investigation of Minichannel Heat Sink for Electronics Cooling”. A CFD Analysis” authored by Waqas Arsad and Hafiz Muhammad Ali.

The first step towards the simulation was designing the required geometries which included two components:

1. Minichannel Heat Sink
2. Three Cartridge Heaters (CSH-201200/120V)

All the designs were made using SOLIDWORKS. The primary design criteria is the heat sink. The other geometry dimensions were taken with reference to the heat sink.

4.1 Minichannel Heat Sink

In SOLIDWORKS the minichannel heat sink was designed first. The parameters required for designing the minichannel heat sink are channel width of heat sink, fin width of heat sink, total width of heat sink, fin height of heat sink, total height of heat sink and total length of heat sink. The values for these parameters were taken with concerning to one of the core research paper we have studied by changing the value in a certain range. The dimensions of these parameters are:

Parameters	Dimension(mm)
W_c	1.00
W_f	0.60
W_t	27.8
H_f	1.00
H_t	50.0
L_t	75.0

Table 4.1 Geometrical Specifications for Minichannel Heat Sink

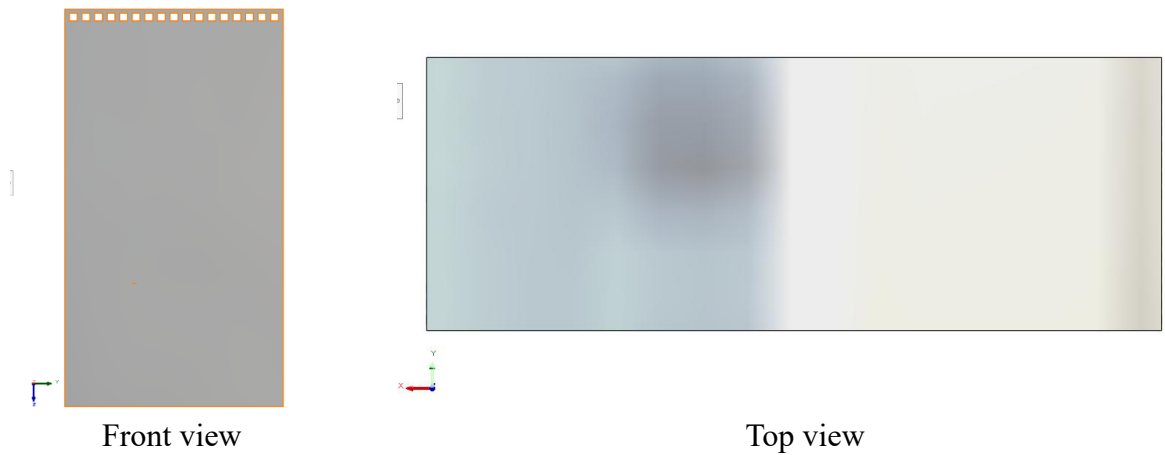


Fig 4.1: Front and Top view of Mini channel Heat Sink

4.2 Cartridge Heaters

After designing the minichannel heat sink we then moved to design the cartridge heaters. For the design of the cartridge heater the dimensions are taken from the reference paper Waqas Arshada, Hafiz Muhammad Ali et.al (2017). In that paper three three cartridge heaters (CSH-201200/120V, Omega, USA) each of maximum 200W were installed. So, the modeled dimensions are taken as the dimension of the CSH-201200/120V.

For design and simulation purposes the parameters taken for cartridge heaters are as follows:

Parameter	Dimensions (mm)
Length	27.8
Height	4.7498
Width	4.7498

Table 4.2: Geometrical Specifications for Cartridge heater

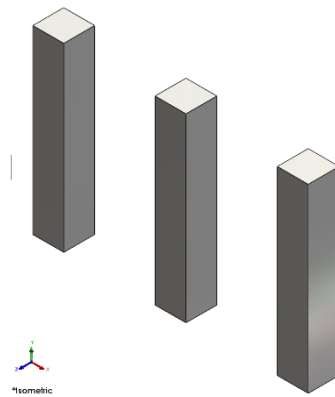


Fig 4.2: Isometric view of three Cartridge heaters

4.3 Heat sink Assembly

Heat sink and three cartridge heaters were assembled. Three cartridge heaters are placed at equal intervals near the minichannels. The design was then converted to IGES format and imported into ANSYS for further analysis.

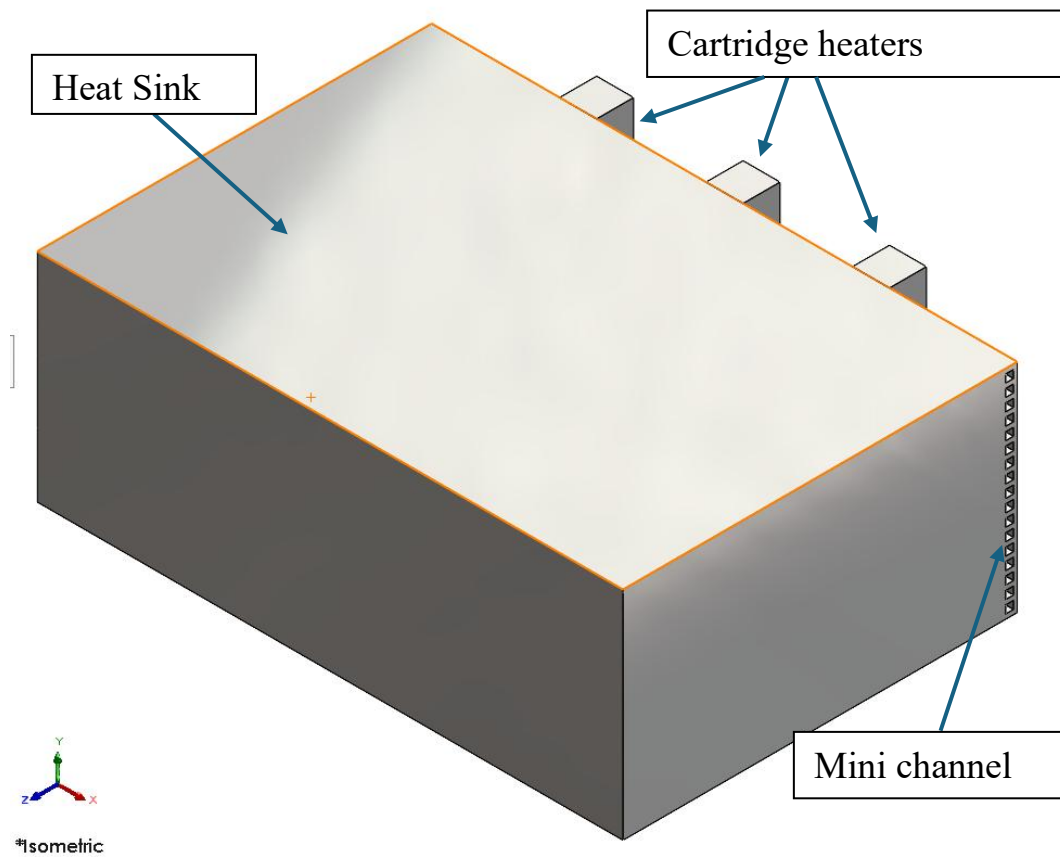


Fig 4.3: Isometric view of Heat sink Assembly

In the analysis phase, ANSYS 2020 R1 version was employed for a thorough investigation. This involved studying fluid flow dynamics, velocity profiles, temperature distributions and pressure gradients.

5.1 Geometry Set-up

Before meshing, the geometry is required to be prepared in SOLIDWORKS. The geometry is imported in IGES format. The geometry may need to be cleaned, repaired or simplified to ensure a robust meshing process.

5.2 Fluid Meshing

After initializing the geometries our task was to perform meshing on these geometries. The meshing process generates a computational mesh that discretizes the geometry of the fluid domain into small cells or elements. The quality and the characteristics of mesh play a critical role in the efficiency and accuracy of the simulation.

ANSYS FLUENT provides us both structured and unstructured meshing methods. The meshing process typically involves the following steps-

a) Surface meshing: The geometry surfaces are meshed by dividing them into smaller elements or faces. ANSYS uses various algorithms to generate surface meshes, such as triangular or quadrilateral elements for 2D problems and tetrahedral or hexahedral elements for 3D problems.

b) Volume meshing: After the generation of the surface mesh, volume mesh is generated by extending the surface elements into interior of the fluid domain.

This step creates 3D elements such as tetrahedra, hexahedra or polyhedral to fill the volume.

c) Mesh Refinement: Mesh refinement techniques are used to improve the mesh quality and resolution in regions of interest. These regions may include boundary layers, areas with complex flow pattern or the regions where high accuracy is desired.

d) Mesh Quality Check: Once the mesh is generated, it undergoes quality checks to ensure it meets certain criteria for accuracy and convergence. Mesh quality is important to achieve accurate and stable simulations.

e) Mesh control and sizing: ANSYS allows the control of resolution of the mesh in different regions of the domain. This enables the users to refine or coarsen the mesh in specific areas to achieve the desired accuracy for computational efficiency.

5.2.1 Mesh Parameters

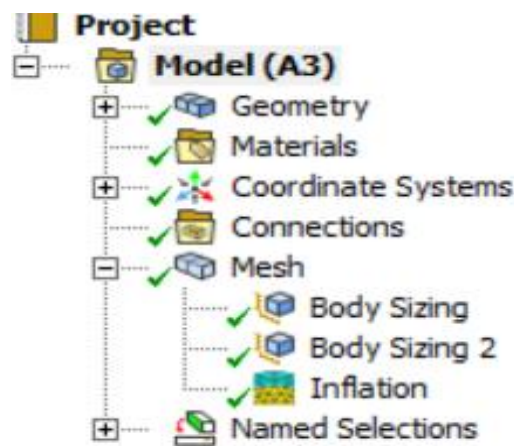


Fig 5.1: Mesh Parameter Specifications

The selected mesh parameters ensure a high-quality mesh that balances accuracy and computational efficiency

- A patch conforming tetrahedral mesh was generated for the fluid domain to accommodate the complex minichannel geometry.
- Inflation layers accurately resolve boundary layer behaviour, critical for predicting heat transfer and fluid flow near walls.
- Body sizing and edge sizing provide mesh refinement in critical regions, ensuring accurate capture of temperature gradients and velocity fields.

Patch conforming tetrahedral mesh adapts to complex geometries like tubes, ensuring the mesh conforms to the surface boundaries.

5.2.2 Mesh Data

The following table depicts the mesh data taken for our minichannel heat sink and cartridge heaters.

Domain	Nodes	Elements
Heat sink	361080	322420
Fluid element	33265	45100
Cartridge Heater	5104	3990
All Domains	892451	1100859

Table 5.1: Mesh Data

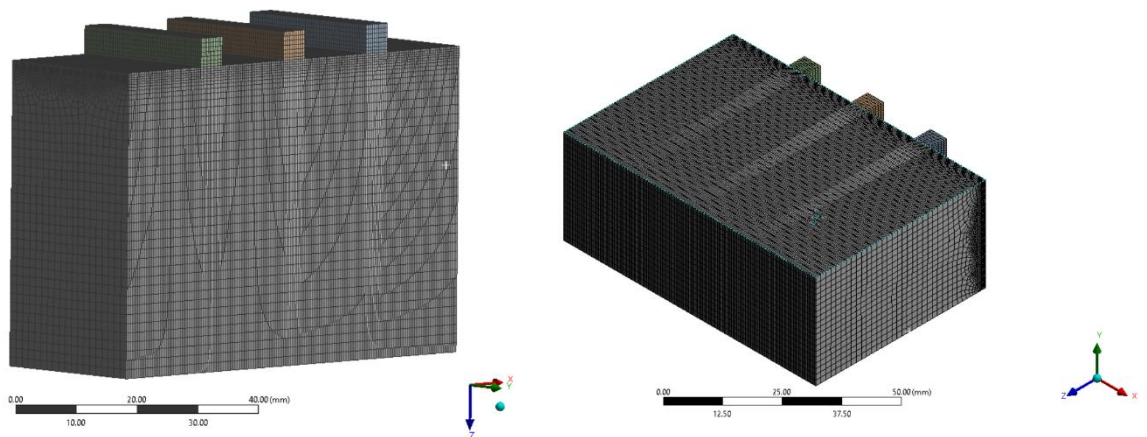


Fig 5.2: Generated Mesh for Present study

5.3 Set-up in FLUENT

a) **Launch Fluent:** Opened the Fluent module and imported the meshed file.

b) **General Setting:** The general settings were defined as follows-

1. Solver setting: Pressure based solver (default)
2. Initial set up: Choose steady for the initial set up.
3. Enabled the energy equation for heat transfer problem.

Solver Type

1. Pressure-Based Solver:

- The pressure-based solver is selected for this simulation because it is suitable for incompressible and mildly compressible flows. Heat sinks typically involve liquid or low-speed liquids flows, where density variations due to pressure are negligible.

Time Settings

2. Steady-State Simulation:

- The steady time setting is chosen because the heat exchanger problem assumes a constant flow rate and steady thermal boundary conditions.

Velocity Formulation

3. Absolute Velocity Formulation:

- Absolute velocity is selected as the velocity formulation because the heat exchanger involves stationary components (e.g., shell, tubes).
- This setting ensures that all velocities are defined relative to a stationary reference frame, which simplifies the analysis.

Gravity

4. Gravitational Acceleration:

- Gravity is disabled in the Y-direction which was account for the gravitational effect on the fluid flow.
- In this heat sink simulation, gravity doesn't play an important role in forced convection effects, particularly for fluids with high density.

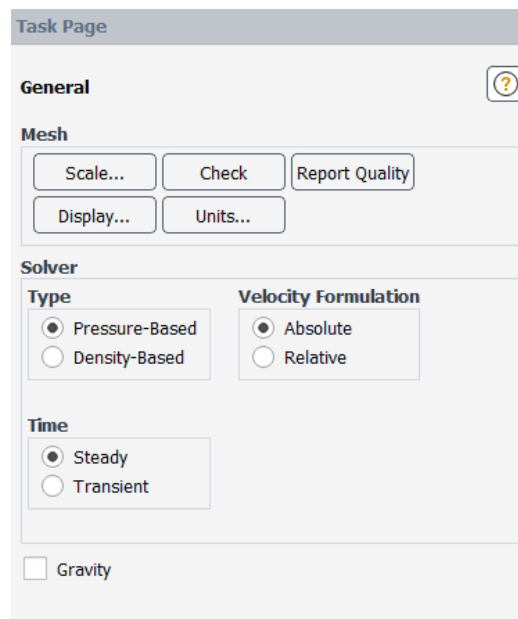
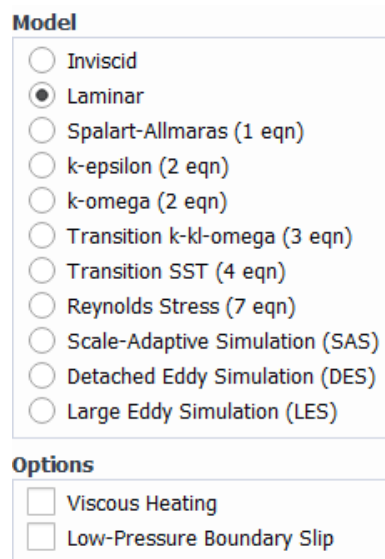


Fig 5.3: General Parameters Setup in ANSYS

By solving the energy equation, we ensure that all conservation laws (mass, momentum, and energy) are satisfied, leading to a robust and reliable CFD simulation. This approach ensures that the temperature distribution and heat transfer characteristics are modelled accurately.

c) Model Selection:



Model

- ☐ Inviscid
- ☒ Laminar
- ☐ Spalart-Allmaras (1 eqn)
- ☐ k-epsilon (2 eqn)
- ☐ k-omega (2 eqn)
- ☐ Transition k-kl-omega (3 eqn)
- ☐ Transition SST (4 eqn)
- ☐ Reynolds Stress (7 eqn)
- ☐ Scale-Adaptive Simulation (SAS)
- ☐ Detached Eddy Simulation (DES)
- ☐ Large Eddy Simulation (LES)

Options

- ☐ Viscous Heating
- ☐ Low-Pressure Boundary Slip

Fig 5.4: Model Parameter Specifications

The Laminar viscous model was chosen for the simulation because the flow is expected to remain below the critical Reynold number. This model assumes that the fluid flows in parallel layers with no disruption between them. It is suitable for natural convection or low-velocity forced convection scenarios where turbulence effects are negligible.

d) Material Selection:

1. For fluid- water liquid
2. Solid- copper and steel

Fluid Materials-Water-Liquid:

- Water is chosen as the working fluid due to its excellent thermal properties, including high heat capacity and thermal conductivity.
- Using “water-liquid” as the material allows Fluent to utilize predefined thermophysical properties like density, specific heat, viscosity, and thermal conductivity, simplifying the setup process.

Solid Materials-Copper:

- Copper is chosen as the solid material for the heat sink due to its superior thermal conductivity.
- Steel is chosen as the solid material for the three cartridge heaters.



Fig 5.5: Selection of Materials

e) Cell Zone Condition: Specify the cell zone conditions for various domains. The Fluid and Solid zones are essential for setting up the computational domain in ANSYS Fluent for simulations involving conjugate heat transfer.

1) Fluid Zones: Fluid Elements

The computational domain was divided into fluid and solid zones to simulate conjugate heat transfer. The fluid zone (fluid elements) was assigned as liquid water to represent the coolant flow. These zones are defined to model:

- i) **Thermal Gradients:** The temperature difference between the inlet and outlet of the fluid.
- ii) **Convective Heat Transfer:** Enables the simulation of heat exchange through fluid motion.
- iii) **Flow Interaction:** Captures the behaviour of fluids as they interact with solid boundaries.

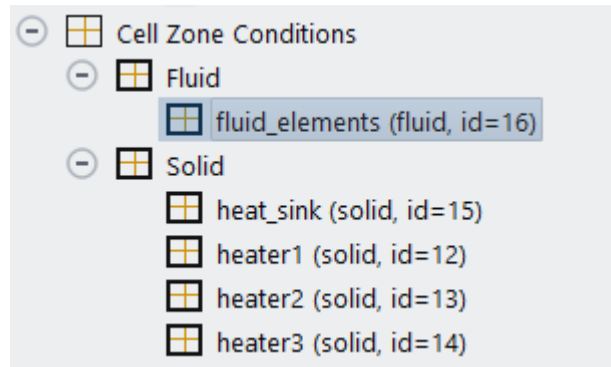


Fig 5.6: Fluid and Solid Zone specifications

2) Solid Zones: Heat sink, Heater1, Heater2 and Heater3

The solid zones were defined for the Heat Sink and the three Heaters (heater1, heater2, and heater3). The heat sink was assigned the material properties of copper, while the heaters were modeled as solid heat sources. This distinction allows the solver to calculate thermal conduction within the solid components and convective heat transfer at the fluid-solid interfaces. Their roles in the simulation are as follows:

- i. Heat sink: Absorbs heat energy from the three cartridge heaters.
- ii. Heaters: Produces a steady heat.

The generation (in Watts/m³) in the heaters is taken as 53142241.84 W/m³. We have obtained this value by equally dividing a total of 100 Watts for each heater, which comes around 33.33 Watts and then dividing this value (33.33 Watts) by the volume of the heater, where volume of the heater is $6.2718 \times 10^{-7} \text{ m}^3$.

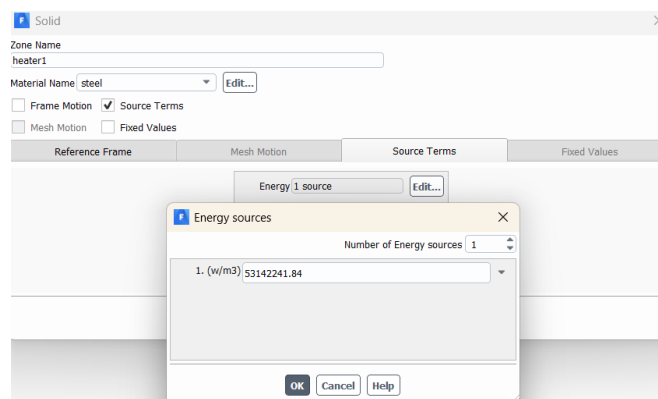


Fig 5.7: Cell zone condition for Heater

f) Boundary Condition: Here we need to assign the inlet boundary conditions for the fluid elements.

Fig 5.8: Wall boundary condition at the inlet of fluid element for Re=300

The above depicts the inlet boundary conditions specified for the fluid elements. The parameters defined here specify the flow behaviour at the inlet, ensuring accurate modeling of the fluid entering the computational domain.

We have inserted four different values of inlet velocities based on the Reynolds number 300, 450, 600 and 750 respectively. Hence, we have done four parallel distinct simulation for each Reynolds number by putting their respective values of different velocities. We have obtained different values of inlet velocities by using the formula

$$Re = \frac{\rho v D}{\mu}$$

Where Re= Reynolds number, v = velocity of fluid (m/s), D = Hydraulic diameter (m), μ = Dynamic viscosity of water (Pa. s), ρ = Density of water (kg/m³)

Re	μ	D	ρ	v
300	0.001003	0.001	998.2	0.301443
450	0.001003	0.001	998.2	0.452164
600	0.001003	0.001	998.2	0.602885
750	0.001003	0.001	998.2	0.753606

Table 5.2: Different values of inlet velocities for their respective Reynolds number

The Supersonic/Initial Gauge Pressure is set to 0 Pascal, indicating that the flow is subsonic, simplifying boundary condition settings as no additional pressure specifications are required. The Direction Specification Method is defined as Normal to Boundary, meaning that the fluid enters perpendicular to the inlet boundary.

For the simulation purpose, we have to give some boundary conditions after designing the whole geometry. The inlet temperatures of the fluid element taken as around 300K.

The outlet gauge pressure of the fluid element is taken as 0-gauge pressure.

g) Mesh Initialization: Hybrid initialization is performed. After completion of initialization, we will proceed towards the run calculation.

h) Run Calculation Setup: The Run Calculation settings in ANSYS Fluent includes Pseudo Transient Settings, Iteration Parameters, and Solution Processing Options.

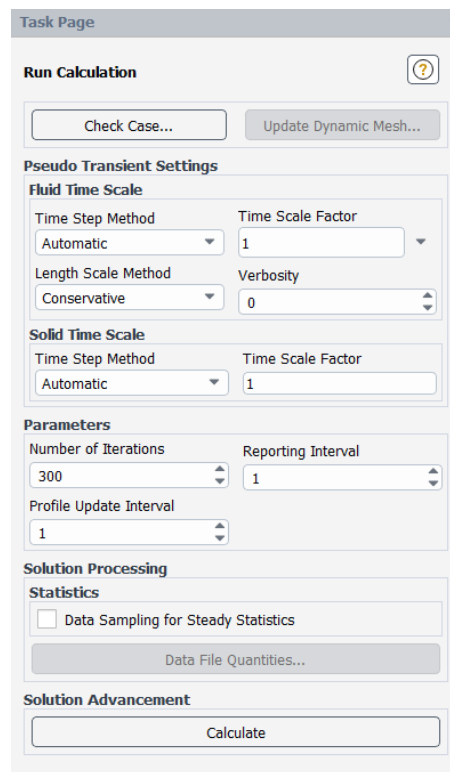


Fig 5.9: Specification of Setup Solver

The Pseudo Transient Settings are activated to improve numerical stability during the simulation. The key parameters include:

1. Fluid Time Scale:

❖ Time Step Method: Automatic

- ◆ The solver automatically adjusts the time step for the fluid flow based on local flow conditions, ensuring stability without requiring manual tuning.

❖ Time Scale Factor: 1

- A factor of 1 maintains a balance between solution speed and accuracy.

❖ Length Scale Method: Conservative

- ◆ This method ensures a stable simulation by adopting a conservative approach for length scale estimation, which is critical for complex geometries like heat exchangers.

2. Solid Time Scale:

❖ Time Step Method: Automatic

- Similar to the fluid settings, the solver automatically adjusts the time step for the solid regions (like walls, tubes, or baffles).

❖ Time Scale Factor: 1

- This ensures the solid region time-stepping aligns with the fluid region, maintaining consistency across conjugate heat transfer calculations.

The pseudo transient approach accelerates convergence for steady-state problems involving strong coupling between fluid flow and heat transfer.

Iteration Parameters

1. Number of Iterations: 300

- This specifies the maximum number of iterations for the solver to converge. A value of 300 is sufficient to allow the simulation to achieve steady-state convergence.

2. Reporting Interval: 1

- This interval ensures that the solution progress is reported after every iteration, allowing for real-time monitoring of residuals and key variables.

3. Profile Update Interval: 1

- The flow and thermal profiles are updated after every iteration, which ensures faster stabilization of the solution variables.

The mathematical equations that have been used for the numerical calculations and modelling are summarized as follows:

When fluid passes, heat transfer occurs between heat sink and coolant. The heat transfer rate is calculated by following Eq. (1)

$$Q = mc_p(T_{out} - T_{in}) \quad (1)$$

All thermo-physical properties are calculated at mean temperature of bulk coolant calculated by following Eq. (2)

$$T_m = \frac{T_{in} + T_{out}}{2} \quad (2)$$

The base temperature is measured 1.5 mm below the wall surface of channel. Fourier's law of heat conduction is used to calculate this temperature given by the following Eq. (3)

$$T_{wl} = T_{tc} - \left(\frac{QH_w}{k_{hs}A_{wl}} \right) \quad (3)$$

Where A_{wl} can be defined as Eq. (4)

$$A_{wl} = w_t L_t \quad (4)$$

The heat transfer rate in term of convective heat transfer coefficient is calculated by Eq. (5),

$$Q = hA_e(LMTD) \quad (5)$$

Where A_e can be defined as the Eq. (6)

$$A_e = tw_c L_t \quad (6)$$

Where total number of channels, $t=17$

LMTD is average temperature difference b/w wall and mean bulk temperature of coolant. This can be calculated by Eq. (7)

$$LMTD = \frac{(T_{wl} - T_{in}) - (T_{wl} - T_{out})}{\ln \left(\frac{T_{wl} - T_{in}}{T_{wl} - T_{out}} \right)} \quad (7)$$

We can get convective heat transfer coefficient Eq. (8) by comparing Eq. (1) & (5)

$$h = \frac{mc_p(T_{out} - T_{in})}{A_e(LMTD)} \quad (8)$$

Convective thermal resistance is important evaluating parameters to define integral-fin efficiency, defined by Eq. (9)

$$R_{th} = \frac{LMTD}{Q} A_c \quad (9)$$

Reynolds number is calculated by Eq. (10),

$$Re = \frac{\rho v_{max} d_h}{\mu} \quad (10)$$

v_{max} , can be calculated by following Eq. (11),

$$v_{max} = \frac{Q_f}{A_c} \quad (11)$$

A_c can be calculated by Eq. (12) respectively,

$$A_c = w_c h_f \quad (12)$$

Hydraulic diameter can be calculated by Eq. (13),

$$d_h = \frac{4A_c}{P_c} \quad (13)$$

Heat transfer performances in term of experimental Nusselt number is defined by Eq. (14)

$$Nu = \frac{hd_h}{k} \quad (14)$$

This experimental Nusselt number is compared with the empirical correction given by Peng and Peterson Eq. (15)

$$Nu = 0.1165 \left(\frac{d_h}{w_{cc}} \right)^{0.81} \left(\frac{h_f}{w_c} \right)^{-0.79} Re^{0.62} Pr^{1/3} \quad (15)$$

7.1 Overview

This chapter presents a detailed discussion of the numerical results obtained from the CFD simulation of the rectangular minichannel heat sink subjected to a constant heat input of 100 W. The simulations were carried out for Reynolds numbers ranging from 300 to 750 under laminar flow conditions using water as the working fluid. The key performance indicators analysed include static pressure distribution, velocity magnitude, temperature distribution, pressure drop, Nusselt number, thermal resistance, log mean temperature difference (LMTD), and wall temperature. The present numerical model was then validated by comparing its results with the published data of Waqas Arshada, Hafiz Muhammad Ali et.al (2017). The validation was carried out by taking four values of Reynolds number (300, 450, 600 and 750 respectively) and heat input of 100 W.

7.2 Static Pressure Distribution

The static pressure contour illustrates a monotonic pressure decrease from inlet to outlet, confirming fully developed laminar flow across the minichannel length. The highest pressure is observed at the inlet manifold, while the minimum pressure occurs at the outlet manifold.

As Reynolds number increases, the overall pressure level inside the channel rises due to increased inlet velocity. This behaviour is consistent with classical internal flow theory, where pressure losses increase with flow rate because of higher viscous shear stresses along the channel walls.

The pressure distribution remains uniform across parallel channels, indicating good flow distribution control and symmetric channel design. No recirculation zones or pressure stagnation regions are observed, confirming that the geometry supports stable single-phase flow.

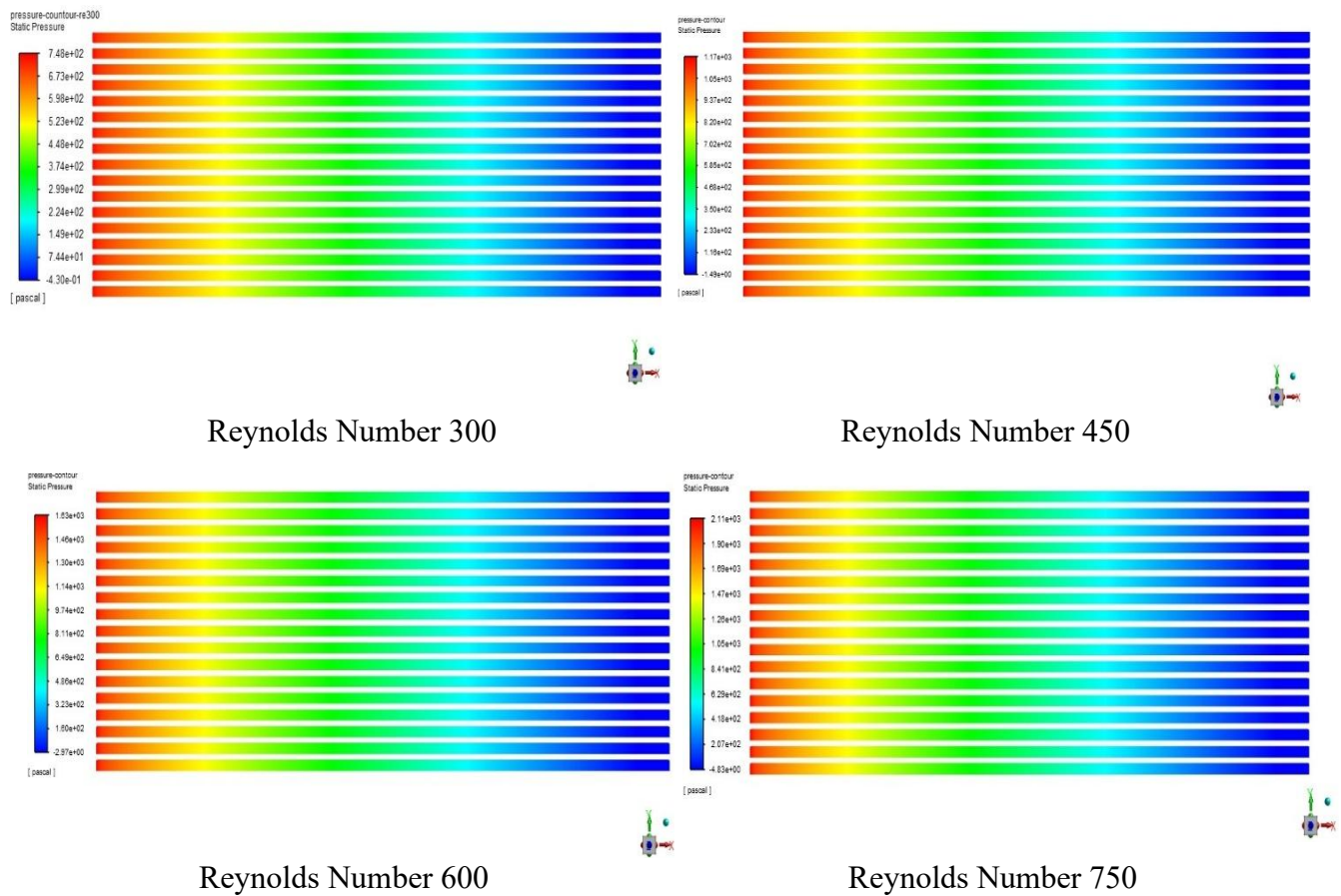


Fig 7.1: Pressure Contours of Fluid Elements for different values of Reynolds Number

In the above pictures we can clearly see that the pressure has the highest values at the inlet, and it decreases gradually along the mini channels. The outlet pressure value has zero-gauge pressure.

Reynolds Number	Pressure Drop (Pa)
300	781.8854
450	1252.519
600	1774.135
750	2345.615

Table 7.1: Values of Pressure Drop for different values of Reynolds Number

7.3 Velocity Distribution

The velocity magnitude contour shows a parabolic velocity profile in each minichannel, characteristic of laminar flow. Maximum velocity occurs at the channel centreline, while velocity reduces to zero at the walls due to the no-slip boundary condition.

With increasing Reynolds number, the peak velocity increases proportionally, leading to thinner hydrodynamic boundary layers. The velocity distribution remains uniform along the channel width, indicating equal flow division among channels, which is crucial for thermal uniformity.

The absence of flow separation or secondary vortices confirms that the heat sink operates entirely in the laminar regime throughout the investigated Reynolds number range.

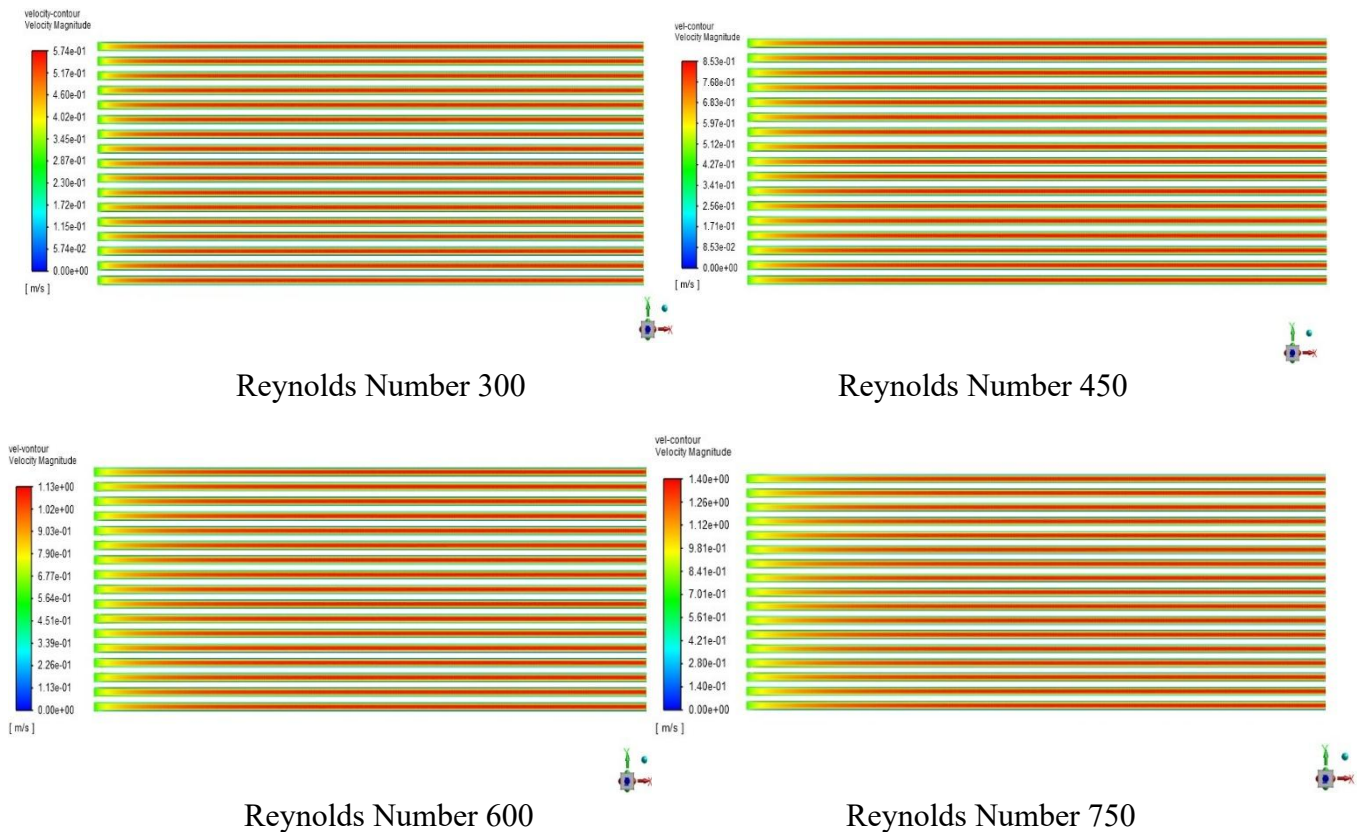


Fig 7.2: Velocity Contours of Fluid Elements for different values of Reynolds Numbers

7.4 Temperature Distribution

The temperature contour clearly demonstrates effective heat removal from the heated base through forced convection. The highest temperatures are observed near the heat source region, while fluid temperature gradually increases along the flow direction.

Distinct temperature jumps are visible at the locations of the cartridge heaters, indicating localized heat addition zones. Downstream of each heater, the coolant absorbs heat, leading to a progressive rise in bulk fluid temperature.

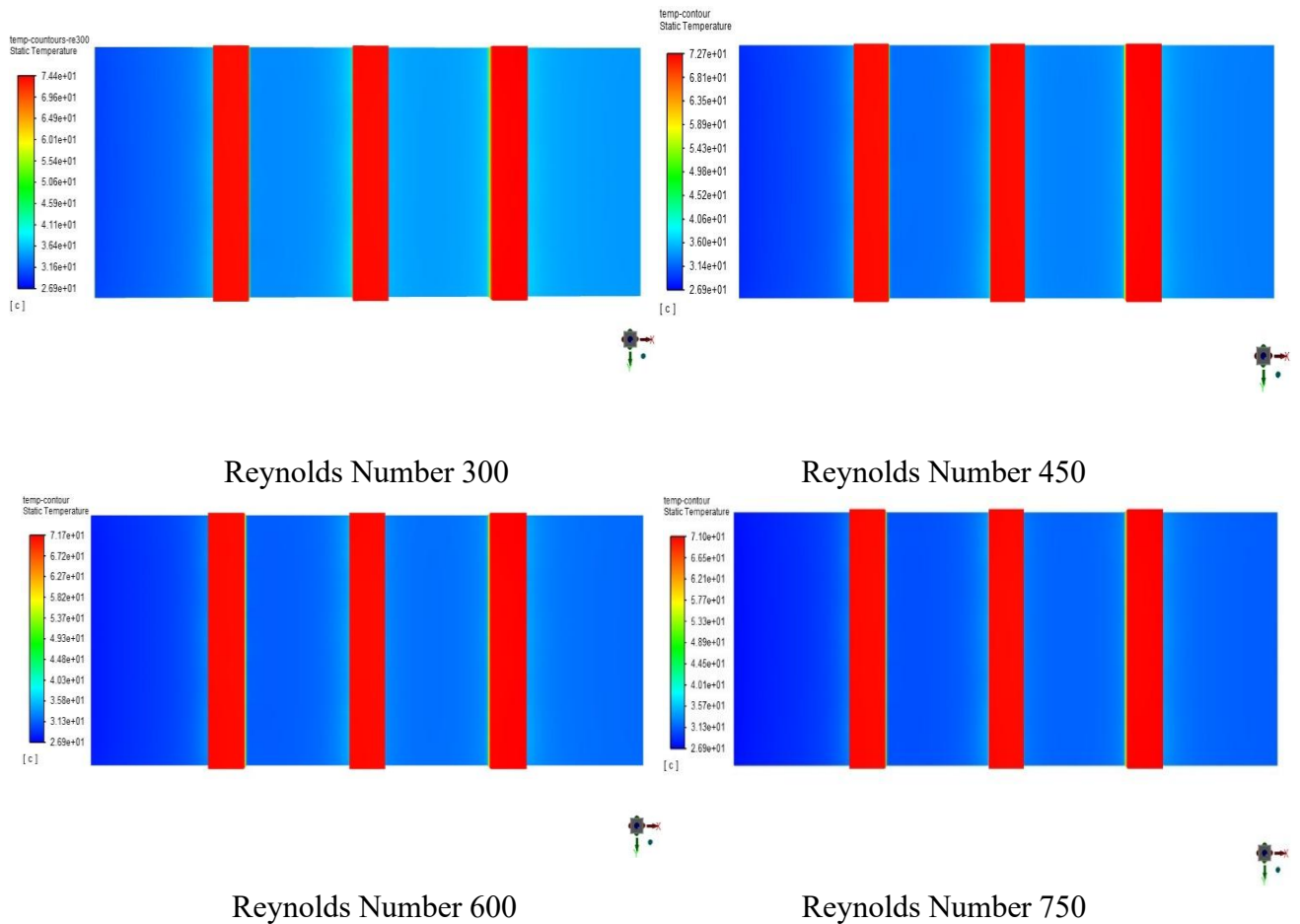


Fig 7.3: Temperature Contours of Minichannel Heat Sink for different values of Reynolds Number

Reynolds Number	Inlet temperature of Fluid (in °C)	Outlet Temperature of Fluid (in °C)
300	27	31.6728
450	27	30.1155
600	27	29.3367
750	27	28.8694

Table 7.2: Inlet and Outlet temperature of Fluid Elements for Different values of Reynolds Number

From the above table we can clearly see that with increase in Reynolds number the temperature of fluid at outlet significantly decreases. The inlet fluid at room temperature absorbs the heat from the heat sink and decreases the heat sink temperature.

As Reynolds number increases, the maximum temperature inside the heat sink decreases significantly. This is attributed to:

- i. Higher convective heat transfer coefficient
- ii. Reduces thermal boundary layer thickness
- iii. Increased heat removal capacity of the coolant

7.5 Pressure Drop Variations with Reynolds Number

The variation of pressure drop with Reynolds number shows an almost linear increasing trend. At lower Reynolds numbers, pressure drop is relatively small, indicating low pumping power requirements. However, as Reynolds number increases, pressure drop rises sharply due to increased frictional losses.

This behaviour highlights the classical trade-off between thermal performance in minichannel heat sinks. While higher Reynolds numbers improve heat transfer, they also demand higher pumping power.

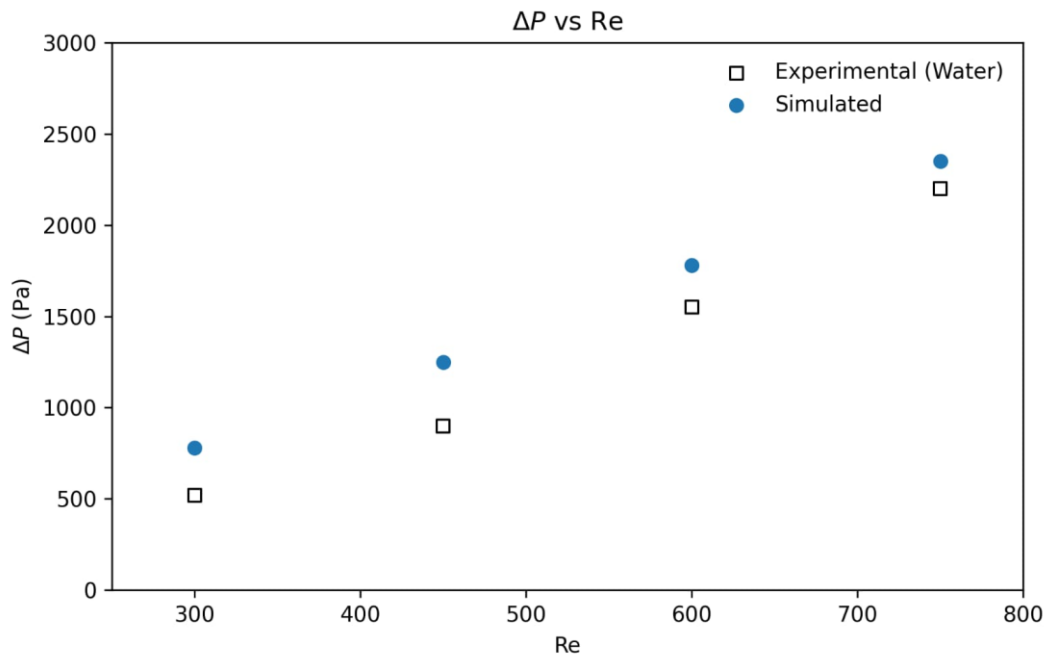


Fig 7.4: Comparison of simulated and Hafiz Muhammad Ali et.al (2017) paper's data for variation of Pressure Drop with respect to Reynolds number

Reynolds Number	Pressure Drop (Pa)
300	781.8854
450	1252.519
600	1774.135
750	2345.615

Table 7.3: Simulated data of variation of Pressure Drop for different values of Reynolds number

7.6 Nusselt Number Variation

The Nusselt number increases monotonically with Reynolds number, indicating enhanced convective heat transfer. This enhancement is primarily due to:

- i) Increased fluid velocity
- ii) Reduced thermal boundary layer thickness
- iii) Improved mixing for the channel walls

The continuous increase in Nusselt number confirms that the flow remains thermally developing throughout the channel length, which is advantageous for heat transfer enhancement in short minichannels.

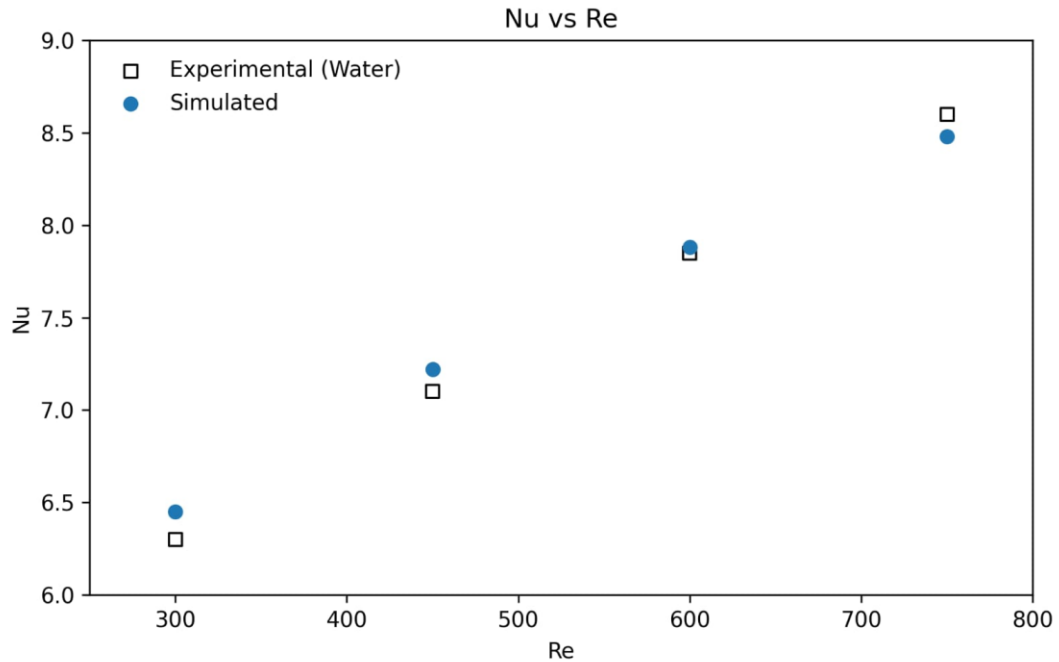


Fig 7.5: Comparison of simulated and Hafiz Muhammad Ali et.al (2017) paper's data for variation of Nusselt number with respect to Reynolds number

Reynolds number	Nusselt number
300	6.4548
450	7.2095
600	7.8815
750	8.4850

Table 7.4: Simulated data of variation of Nusselt number for different values of Reynolds number

7.7 Thermal Resistance Analysis

Thermal resistance decreases significantly with increasing Reynolds number. At low Reynolds numbers, heat removal is limited by weak convection, resulting in higher thermal resistance. As Reynolds number increases, improved convection reduces the temperature difference between the wall and fluid.

The reduction in thermal resistance demonstrates that the minichannel heat sink becomes more effective at higher flow rates. However, the marginal reduction in thermal resistance diminishes at higher Reynolds numbers, indicating the onset of diminishing returns.

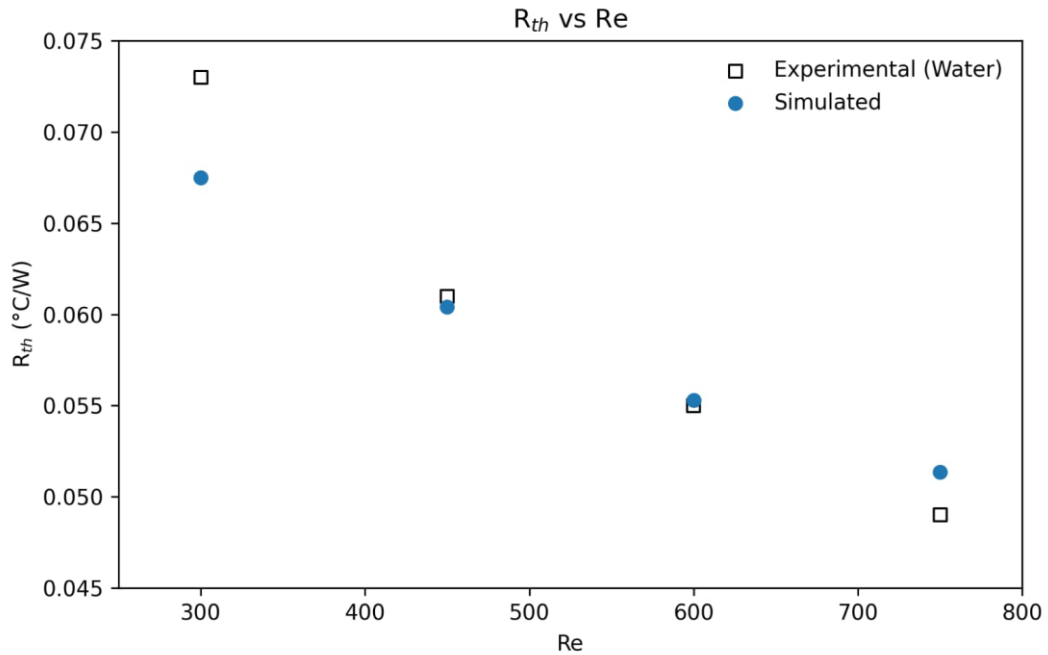


Fig 7.6: Comparison of simulated and Hafiz Muhammad Ali et.al (2017) Paper's data for variation of Thermal Resistance with respect to Reynolds number

Reynolds number	Thermal Resistance (°C/W)
300	0.0675
450	0.0604
600	0.0553
750	0.0514

Table 7.5: Simulated data of variation of Thermal Resistance for different values of Reynolds number

7.8 Log Mean Temperature Difference

The LMTD decreases with increasing Reynolds number. At lower flow rates, larger temperature gradients exist between the wall and coolant, resulting in higher LMTD. As Reynolds number increases, enhanced convection reduces this temperature difference.

This reduction in LMTD indicates more efficient heat extraction and improved thermal uniformity along the channel length.

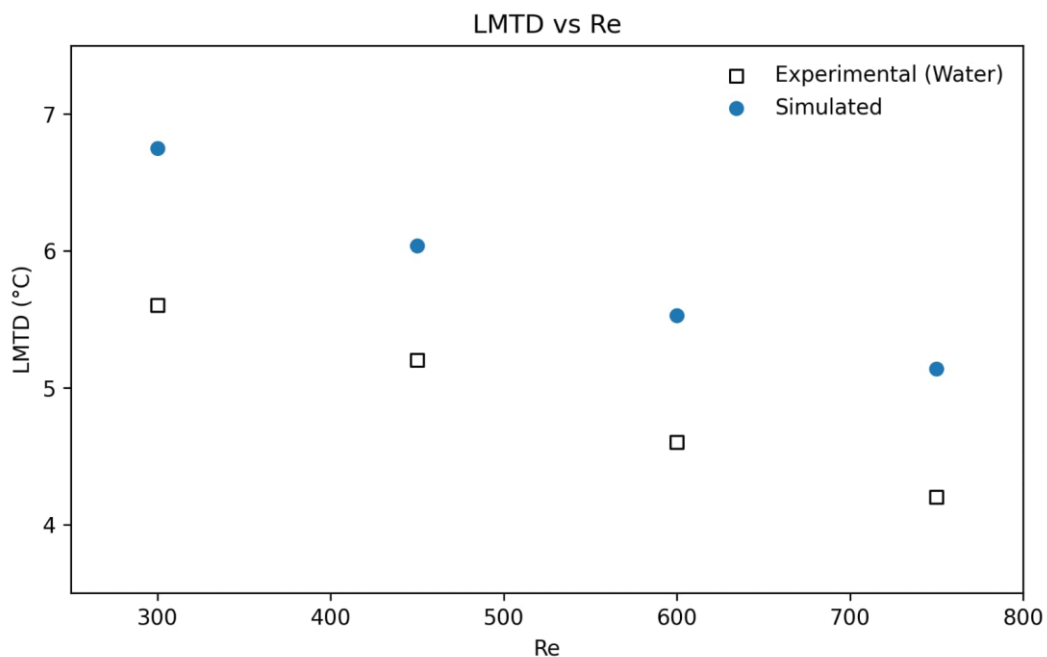


Fig 7.7: Comparison of simulated and Hafiz Muhammad Ali et.al (2017) Paper's data for variation of LMTD with respect to Reynolds number

Reynolds number	LMTD (°C)
300	6.750
450	6.044
600	5.529
750	5.135

Table 7.6: Simulated data of variation of LMTD for different values of Reynolds number

7.9 Wall Temperature Variation

The average wall temperature (which is located below 1.5 mm from the fluid element surface) decreases steadily with increasing Reynolds number. This behaviour confirms that higher coolant flow rates are effective in suppressing wall temperature rise.

Maintaining a low wall temperature is critical for electronics cooling applications, as excessive temperatures can cause thermal stress, reliability issues, and performance degradation.

The wall temperature variation also shows uniformity across the heat sink surface, confirming that the design effectively avoids localized hot spots.

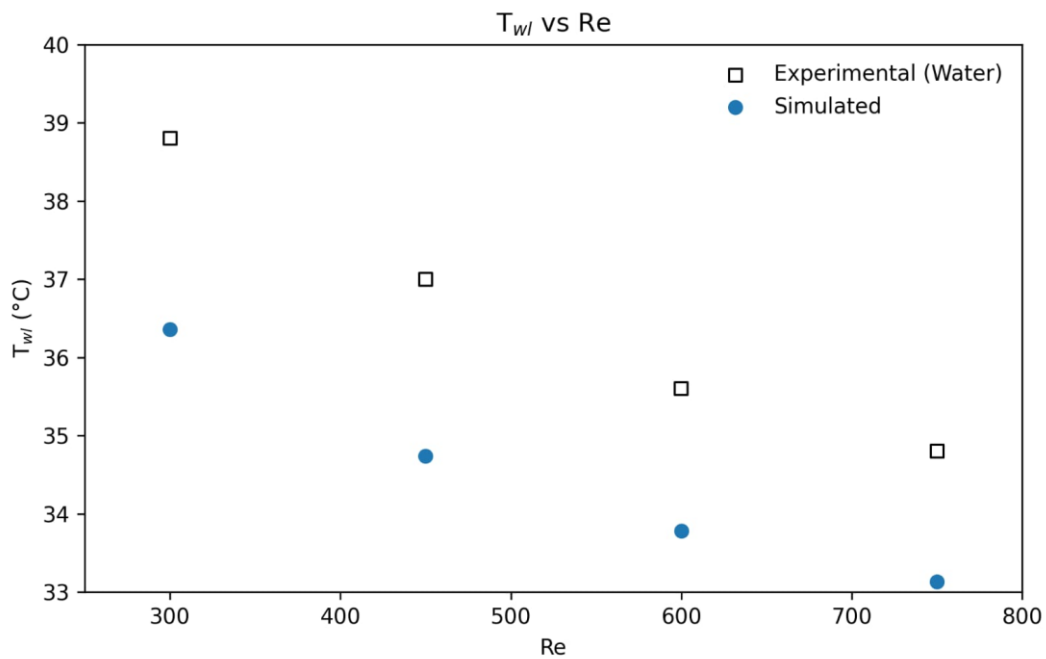


Fig 7.8: Comparison of simulated and Hafiz Muhammad Ali et.al (2017) Paper's data for variation of Temperature of base wall with respect to Reynolds number

Reynolds number	Average Wall Temperature (°C)
300	36.3555
450	34.7359
600	33.7799
750	33.1275

Table 7.7: Simulated data of variation of average base wall temperature for different values of Reynolds number

7.10 Overall Performance Assessment

The numerical results confirm that the rectangular minichannel heat sink provides:

- i) High heat transfer efficiency
- ii) Uniform temperature distribution
- iii) Stable laminar flow behaviour
- iv) Predictable pressure drop characteristics

An optimal operating range is observed where thermal resistance reduction is significant without excessive pressure drop. This balance is essential for practical electronics cooling systems, where both thermal and hydraulic performances must be optimized simultaneously.

7.11 Comparing with Published Literature

The observed trends in pressure drop, Nusselt number, and thermal resistance are consistent with previously published experimental studies on minichannel heat sink published by Waqas Arshada, Hafiz Muhammad Ali et.al (2017). The laminar flow behaviour, monotonic increase in Nusselt number, and decreasing thermal resistance with Reynolds number align well with established findings reported in literature.

8.1 CONCLUSION

In this project, a numerical investigation of a minichannel heat sink for electronics cooling was carried out to analyse its thermal and hydraulic performance. This study focused on understanding the heat transfer characteristics, temperature distribution, and pressure drop within the minichannel heat sink under specified operating conditions. Numerical simulations were performed using appropriate CFD techniques, enabling detailed visualization of flow behaviour and thermal fields.

The results demonstrate that minichannel heat sinks are highly effective in removing high heat fluxes generated by modern electronic devices. Due to the large surface-area-to-volume ratio of minichannels, enhanced convective heat transfer was observed, leading to a significant reduction in maximum substrate temperature. The study also revealed that increasing coolant flow rate improves heat transfer performance, though it results in a higher pressure drop, indicating a trade-off between thermal efficiency and pumping power. The numerical approach adopted in this work proved to be a reliable and cost-effective method for evaluating minichannel heat sink performance prior to experimental implementation.

Overall, the findings confirm that minichannel heat sinks offer a compact, efficient, and reliable solution for thermal management of electronic components, making them suitable for high-performance and miniaturized electronic systems.

8.2 FUTURE SCOPE

The numerical investigation of a minichannel heat sink presented in this project provides a strong foundation for further research and development in the field of advanced electronics cooling. However, with the continuous miniaturization of electronic components and the increasing heat flux generated by modern devices, there exists a wide scope for future enhancement and exploration.

One important extension of the present work is the use of advanced working fluids. Future studies can investigate the performance of nanofluids, hybrid nanofluids, and phase change materials (PCMs) as coolants in microchannels. The inclusion of nanoparticles such as Al_2O_3 , CuO , or graphene can significantly enhance thermal conductivity, thereby improving heat transfer performance without substantial modification to the heat sink design.

Another major area of future research is two-phase flow and boiling heat transfer in minichannels. While the current study focuses on single-phase flow, two-phase cooling techniques can handle extremely high heat fluxes and are highly relevant for high-power electronics, laser systems, and data centres. Modeling boiling phenomena, flow instability, and vapor–liquid interactions can lead to more efficient thermal management solutions.

Geometrical optimization of microchannels also offers significant scope for improvement. Future work can explore different microchannel shapes such as trapezoidal, triangular, wavy, or pin-fin integrated channels. Optimization techniques including genetic algorithms, response surface methodology, and artificial intelligence-based approaches can be employed to achieve optimal performance in terms of maximum heat transfer and minimum pressure drop.

The study can further be extended to include transient thermal analysis, which would help in understanding the dynamic response of microchannel heat sinks under varying heat loads and operating conditions. This is particularly important for electronic devices that operate under fluctuating power inputs.

Experimental validation of the numerical results is another important future direction. Fabrication of microchannel heat sinks using MEMS technology or advanced additive manufacturing methods can help in validating simulation outcomes and improving model accuracy.

Finally, integration of microchannel heat sinks with modern electronic packaging systems, along with an assessment of energy efficiency, pumping power requirements, and environmental sustainability, can make the technology more viable for real-world industrial applications.

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